# Empirical Methods for Policy Evaluation

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# Shift-Share Instrumental Variables and Models of Firm Dynamics

- Shift-Share Designs
  - $\Rightarrow$ Instrumental variables under heterogenous treatment effects
  - $\Rightarrow$ Shift-share design as leveraging a shift-level natural experiment
  - $\Rightarrow$ Shift-share design as combining difference-in-differences
- Rural migration and urban informality
  - $\Rightarrow$ Imbert and Ulyssea (ECMA, 2026)

# Instrumental variables under heterogenous treatment effects

# Recap on Instrumental Variables (IV)

- Recall from the intro slides

$$\begin{aligned}\text{ATE} &= \mathbb{E}(Y_i^1 - Y_i^0) \\ &= \mathbb{E}(Y_i^1 \mid D_i = 1) - \mathbb{E}(Y_i^0 \mid D_i = 0) \\ &= \underbrace{\mathbb{E}(Y_i^1 - Y_i^0 \mid D_i = 1)}_{\text{ATT}} + \underbrace{\mathbb{E}(Y_i^0 \mid D_i = 1) - \mathbb{E}(Y_i^0 \mid D_i = 0)}_{\text{Selection bias}}\end{aligned}$$

- A standard solution is to find $Z_i$ such that

$$\mathbb{E}(Y_i^0 \mid Z_i, D_i = 1) = \mathbb{E}(Y_i^0 \mid Z_i, D_i = 0)$$

# Potential and Observed Treatments and Outcomes

- Assume binary instrument (offered treatment), $Z_i \in \{0, 1\}$
- Potential treatments are $D_i(z)$ and realized treatment

$$D_i = (1 - Z_i)D_i(0) + Z_i D_i(1)$$

- Potential outcomes are $Y_i(z, d)$ and realized outcomes

$$Y_i = Y_i(Z_i, D_i(Z_i)) = \begin{cases} Y_i(0,0) & \text{if } Z_i = 0, D_i(0) = 0 \\ Y_i(0,1) & \text{if } Z_i = 0, D_i(0) = 1 \\ Y_i(1,0) & \text{if } Z_i = 1, D_i(1) = 0 \\ Y_i(1,1) & \text{if } Z_i = 1, D_i(1) = 1 \end{cases}$$

# IV Assumptions

A1 Relevance (first stage)

$$\mathbb{E}[D_i(1)] \neq \mathbb{E}[D_i(0)]$$

A2 Exclusion Restriction (no direct effect of the offered treatment on outcome)

$$Y_i(0,d) = Y_i(1,d) = Y_i(d), \forall d \in \{0,1\}$$

A3 Full Exogeneity ($Z_i$ is exogenous with respect to both the outcome and the treatment)

$$Z_i \perp (D_i(0), D_i(1), Y_i(0), Y_i(1))$$

A4 Monotonicity ($Z_i$ weakly increases incentives to take the treatment)

$$D_i(1) \geq D_i(0), \forall i$$

# Local Average Treatment Effects (LATE)

- Under A1-A4 we can identify the ATE for compliers (LATE)

$$\tau^{\text{late}} = \frac{\frac{1}{N}\sum_{i=1}^{N}[Y_i(D_i(1)) - Y_i(D_i(0))]}{\frac{1}{N}\sum_{i=1}^{N}[D_i(1) - D_i(0)]}$$

  $\Rightarrow$ Do not confuse LATE with naive estimand: units who do not comply with offered treat are dropped

$$\tau^{\text{naive}} = \frac{1}{N_c}\sum_{i:D_i(0)=0, D_i(1)=1}[Y_i(1) - Y_i(0)]$$

- Common practice is 2SLS regression of $Y_i$ on $D_i$ using $Z_i$ as the excluded instrument

# LATE Example: The Vietnam draft lottery

**Table 24.1.** ***Summary Statistics for the Angrist Draft Lottery Data***

| | Non-Veterans ($N_c = 6{,}675$) | | | | Veterans ($N_t = 2{,}030$) | | | |
|---|---|---|---|---|---|---|---|---|
| | Min | Max | Mean | (S.D.) | Min | Max | Mean | (S.D.) |
| Draft eligible | 0 | 1 | 0.24 | (0.43) | 0 | 1 | 0.40 | (0.49) |
| Yearly earnings (in $1,000's) | 0 | 62.8 | 11.8 | (11.5) | 0 | 50.7 | 11.7 | (11.8) |
| Earnings positive | 0 | 1 | 0.88 | (0.32) | 0 | 1 | 0.91 | (0.29) |
| Year of birth | 50 | 52 | 51.1 | (0.8) | 50 | 52 | 50.9 | (0.8) |

- $\widehat{\tau}^{\text{itt}} = -0.213$ $(\widehat{s.e.} = 0.20)$
- $\widehat{\tau}^{\text{fs}} = 0.1460$ $(\widehat{s.e.} = 0.0108)$
- $\widehat{\tau}^{\text{2sls}} = \frac{\widehat{\tau}^{\text{itt}}}{\widehat{\tau}^{\text{fs}}} = -\frac{0.21}{0.1460} = -1.46$ $(\widehat{s.e.} = 1.36)$

# Possible violations of the exclusion restriction

- Never takers: $Y_i(0,0) \neq Y_i(1,0)$

  $\Rightarrow$ Dodging the draft if assigned will likely involve differences in earnings

- Always takers: $Y_i(0,1) \neq Y_i(1,1)$

  $\Rightarrow$ If accepting means different tasks, then possible differences in later earnings

- Compliers and defiers: $Y_i(0,d) \neq Y_i(1,d)$

  $\Rightarrow$ Effect on earnings is attributed to serving in the military and not to the draft itself

# Possible violations of the monotonicity assumption

- In the Vietnam draft lottery example
  - $\Rightarrow$ $D_i(1) < D_i(0)$ if some would be willing to volunteer if not drafted but resist to serve if drafted
- The effect of fertility on female labor supply ($Z_i$ is sex composition of siblings)
  - $\Rightarrow$ $D_i(1) \geq D_i(0)$ if all families have same preference for sex mix
  - $\Rightarrow$ $D_i(1) < D_i(0)$ if there are families with stopping rule for two male children
- Judge-leniency design ($Z_i$ randomly assigns judges to cases)
  - $\Rightarrow$ $D_i(1) \geq D_i(0)$ if judge A is stricter than B then A always convicts whenever B would convict
  - $\Rightarrow$ $D_i(1) < D_i(0)$ whenever the two judges disagree

# 2SLS $\neq$ LATE

- With covariates, 2SLS estimator reflects treatment effects for compliers and always/never-takers
  - $\Rightarrow$ LATE interpretation holds only when controlling for covariates non-parametrically
- If $Z_i \in \{1, .., K\}$ then $K$ compliers' group, one for each subsequent pair of values of $Z_i$
  - $\Rightarrow$ LATE $= \sum_{k=1}^{K} \omega_k \text{LATE}_k$, where $\omega_k$ depend on the instrument
  - $\Rightarrow$ Same intuition for when $Z_i$ is a vector (multiple instruments)
- With unordered, non-cardinal $Z_i$ more assumptions on treatment choice behavior are needed
  - $\Rightarrow$ E.g., Average Monotonicity ($\omega_k \geq 0$, $\forall k$), Unordered Partial Monotonicity

# Estimating Target Parameters using a Selection Model

- Consider a threshold crossing model under IV assumptions and covariates $X_i$

$$D_i = \mathbb{I}(V_i \leq \nu(Z_i, X_i))$$

  $\Rightarrow$ $V_i$ is a random variable that captures $i$'s unobserved tendency to take treatment

- Given full exogeneity of $Z_i$, probability of treatment (propensity score) is

$$P(D_i|Z_i = z, X_i = x) = P(V_i = \nu(z, x)|X_i = x) \equiv F_{V|X}(\nu(z, x)|x) \equiv U_i \sim U[0, 1]$$

  $\Rightarrow$ $U_i$ is the quantile of resistance to treatment: $D_i = \mathbb{I}(U_i \leq P(Z_i, X_i))$

# Marginal Treatment Effect (MTE)

- The MTE is the ATE among subpopulations with the same propensity to take treatment

$$MTE(u, x) = \mathbb{E}(Y_i(1) - Y_i(0)|U_i = u, X_i = x)$$

- Any target parameter can be written as a weighted average of MTE. E.g:

$$ATE(x) = \int_0^1 MTE(u, x)du$$

$$LATE(x, u', u) = \frac{1}{u' - u} \int_{u'}^{u} MTE(u, x)du$$

$\Rightarrow$ Where $u' = P(D_i(0) = 1)$ and $u = P(D_i(1) = 1)$

# MTE Identification

- The derivative of the outcome wrt the propensity score yields the MTE evaluated at $P(Z_i, X_i) = u$

$$MTE(x,p) = \frac{\partial \mathbb{E}(Y_i|X_i = x, P(Z_i, X_i) = p)}{\partial p}$$

  $\Rightarrow$ At $P_i = p$ those with $u < p$ are treated and those with $u = p$ are indifferent

  $\Rightarrow$ Marginally increasing $p$ by $dp$ shifts indifferent individuals into treatment

  $\Rightarrow$ The associated increase in $Y_i$ is $dY = dp \times MTE(x,p)$

- The Local IV estimator is

$$\tau^{\text{liv}} = \frac{\mathbb{E}(Y_i|X_i = x, P(Z_i, X_i) = p(z', x)) - \mathbb{E}(Y_i|X_i = x, P(Z_i, X_i) = p(z, x))}{\mathbb{E}(D_i|X_i = x, Z_i = z') - \mathbb{E}(D_i|X_i = x, Z_i = z)}$$

# MTE Identification

- $\tau^{\text{liv}}$ needs continuous variation in $P_i(Z_i, X_i)$ around $u$, conditional on $X_i = x$
- Hence, it requires continuous variation in $Z_i$ (combine multiple IVs to extend support of $P_i$)
- Continuous covariate variation can be used as a substitute if the MTE is assumed to be separable

$$MTE(u, x) = m_X(x) + m_U(u) = X_i(\beta(1) - \beta(0)) + \mathbb{E}(U(1) - U(0)|U = p)$$

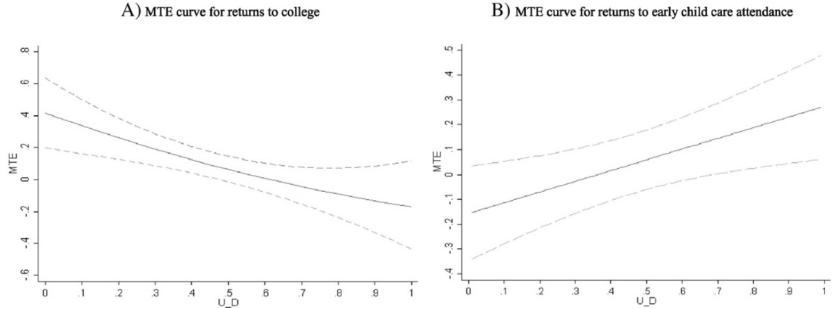
  $\Rightarrow$ $X_i$ affects the intercept of MTE curve but not its slope

  $\Rightarrow$ Identification relies on joint distribution of $P_i(Z_i, X_i)$ rather than conditional $P_i(Z_i|X_i)$

# MTE Estimation

- Estimate the propensity score (e.g. Probit or Logit)
- Regress $Y$ on $D$, $X$, $D \times X \times \hat{p}$, and a polynomial in $\hat{p}$ of degree $k$
- Calculate the derivative of $Y$ with respect to $\hat{p}$ at given values of $X$
- Plot the MTE curve over $U$ and bootstrapped CIs
- Aggregate the MTE over the distribution of covariates with appropriate weights (ATE, ATT, etc.)

# MTE Example: The Returns to Education




| | (1) Returns to college | (2) Returns to early child care attendance |
|---|---|---|
| ATE | 0.067* (0.038) | 0.059 (0.072) |
| TT | 0.143*** (0.035) | −0.051 (0.080) |
| TUT | −0.007 (0.071) | 0.173** (0.085) |
| IV | 0.095** (0.039) | 0.065 (0.133) |

# Shift-Share Instrumental Variables

- General structure of shift-share IV (SSIV)

$$Z_i = \sum_{k=1}^{K} S_{ik} g_k$$

- $g_1, ..., g_K$ are shifts not specific to $i$ (e.g. industry-level employment growth)
- $S_{ik}$ are exposure shares (e.g. initial share of $k$ in $i$'s employment)
- When $\sum_k S_{ik} = 1 \; \forall i$, $Z_i$ is a (share-)weighted average of the shifts

# Shift-Share IV: Decomposition

- $D_i = \frac{X_{i1} - X_{i0}}{X_{i0}}$
- This can be decomposed as $X_{it} = \sum_k X_{ikt}$, hence

$$\frac{X_{i1} - X_{i0}}{X_{i0}} = \sum_k \underbrace{\frac{X_{ik0}}{X_{i0}}}_{\text{Share}} \underbrace{\frac{X_{ik1} - X_{ik0}}{X_{ik0}}}_{\text{Local shift}}$$

- Use a set of common shifts $g_k = \frac{X_{k1} - X_{k0}}{X_{k0}}$ to replace the local shifts

# Shift-Share IV: Example 1 (Bartik, 1991)

- Consider the (inverse) regional supply equation:

$$\Delta \log w_i = \tau \underbrace{\Delta \log L_i}_{= \sum_k S_{ik} \Delta \log L_{ik}} + \underbrace{\epsilon_i}_{\text{labor supply shocks}}$$

- Need a regional labor demand shock as IV

- $Z_i = \sum_k S_{ik} g_k$ by replacing $\Delta \log L_{ik}$ with national industry shift $g_k$
  - $\Rightarrow$ $S_{ik}$ = share of employment in industry $k$
  - $\Rightarrow$ $g_k$ = observed growth rate of industry employment

# Shift-Share IV: Example 2 (Card, 2009)

- Consider the (inverse) elasticity of substitution between native and immigrants

$$\log \frac{w_i^{\text{Immigrant}}}{w_i^{\text{Native}}} = \tau \log \frac{L_i^{\text{Immigrant}}}{L_i^{\text{Native}}} + \underbrace{\epsilon_i}_{\text{change in relative labor demand in region } i}$$

- Need a relative labor supply shock as IV

- $Z_i = \sum_k S_{ik} g_k$: migration prediction from pull & push factors
  - $\Rightarrow$ $S_{ik}$ = initial share of immigrants of origin $k$ in $i$'s population
  - $\Rightarrow$ $g_k$ = observed national migration rate from $k$, normalized by the stock of migrants from $k$ in the US

# The SSIV Exogeneity Challenge

- How should we think about $E[Z_i \epsilon_i] = 0$ ?
  - $\Rightarrow$ Which properties of the shifts and/or shares can make $Z_i$ exogenous ?
  - $\Rightarrow$ Do we need exogeneity of both shifts and shares, or just one ?
- Two approaches + sets of sufficient conditions
  - $\Rightarrow$ Many exogenous shifts: leveraging a shift-level natural experiment
  - $\Rightarrow$ Exogenous shares: pooling diff-in-diffs based on heterogenous exposure shares

# SSIV as leveraging a shift-level natural experiment

# Identification: Intuition

- Suppose shifts are as-good-as-random: $\mathbb{E}[g_k \mid \epsilon, S] = \alpha$
- Then $\mathbb{E}(Z_i) = \mathbb{E}[\sum_k S_{ik} g_k \mid \epsilon, S] = \alpha \sum_k S_{ik}$
  - $\Rightarrow$ If shares add up to one, then $\mathbb{E}(Z_i) =$ constant
- A weighted average of as-good-random shifts is as-good-as random
  - $\Rightarrow$ If industries with high/low $g_k$ are comparable, regions specializing in those industries are comparable
  - $\Rightarrow$ True whatever the initial shares are, we don't need Cov $[S_{ik}, \epsilon_i] = 0$
- Translate random variation across $k$ into random variation across $i$

# Identification: Sufficient conditions

- Using the SSIV decomposition

$$\mathbb{E}\left[\sum_i Z_i \epsilon_i\right] = \mathbb{E}\left[\sum_i \sum_k S_{ik} g_k \epsilon_i\right] = \mathbb{E}\left[\sum_k s_k g_k \bar{\epsilon}_k\right]$$

  $\Rightarrow$ Where $s_k = \sum_i S_{ik}$ and $\bar{\epsilon}_k = \sum_i S_{ik}\epsilon_i / \sum_i S_{ik}$ (exposure-weighted averages of residuals)

- $\frac{1}{N}\sum_i Z_i \epsilon_i \to 0$ if

  A1 $g_k$ are uncorrelated with $\bar{\epsilon}_k = \sum_i S_{ik}\epsilon_i / \sum_i S_{ik}$

  A2 The effective number of shifts is large (otw corr $(g_k, \bar{\epsilon}_k) \neq 0$ by chance)

  A3 Shares add up to one for each observation $i$ (otw $g_k$ vary across $i$ through $\sum_k S_{ik}$)

# Incomplete Shares

- What if $\sum_k S_{ik} \neq 1$, so $Z_i$ is a weighted sum, not a weighted average?
  - $\Rightarrow$ E.g. lottery that randomly assigns subsidies $g_k$ to manufacturing industries $k$
- Regions with more manufacturing will mechanically get higher $Z_i$
- Thus, controlling for $X_i = \sum_k S_{ik}$ avoids bias

# Shift-level Controls

- What if shifts are as-good-as-random only after controlling for some $q_k$?
  - $\Rightarrow$ E.g. industry subsidies are random only controlling for dummy of hi-tech
- $\mathbb{E}(Z_i) = \alpha \sum_k S_{ik}(\alpha_0 + \alpha_1 q_k)$
- To avoid bias, it's sufficient to control for $\sum_k S_{ik}$ and $\sum_k S_{ik} q_k$
- E.g. control for the local share of hi-tech industries
  - $\Rightarrow$ Leverage variation across regions with similar composition of high-tech

# Inference

- Observations with similar shares are exposed to same shocks: $g_k$ and (unobserved) $\nu_k$
- Conventional Liang-Zeger clustering (e.g. by state) wouldn't capture that
  - $\Rightarrow$ Placebo shifts $g_k^\star$ generate significant reduced-form effect in $\sim$50% simulations
- If each region was exposed to just one industry, we'd cluster by industry
- But how to do it in a "many-to-one" case?

# Shift-level ("Exposure-robust") Standard Errors

- Residualize outcomes and treatment on $i$-level controls: $V_i^{\perp}$
- Average across observations with weights $S_{ik}$: $\bar{V}_k^{\perp} = \sum_i S_{ik} V_i^{\perp} / \sum_i S_{ik}$
- SSIV estimate is equal to the estimate from a weighted 2SLS with $g_k$ as IV

$$\bar{Y}_k^{\perp} = \tau \bar{D}_k^{\perp} + \gamma' q_k + \bar{\epsilon}_k$$

  $\Rightarrow$ Where the weights are $s_k = \frac{1}{N} \sum_i S_{ik}$

- Then use robust or clustered SE as in any other regression

# Example: The China Shock (Autor, Dorn, Hanson, 2013)

- ADH study the effects of import competition on local labor market outcomes
- Region $i$ = commuting zone (N=722), industry $k$ = SIC4 manufacturing (K=397)
- Two periods $t$: 1991-2000 and 2000-2007
- $Y_{it}$ = local change in manufacturing employment rate
- $D_{it}$ = local growth of exposure to Chinese imports

$$Z_{it} = \sum_k S^L_{ikt_0} \Delta(\text{Non-US imports from China}_{kt})$$

- Control for period FE and some initial regional characteristics

# The China Shock: Balance Tests

- Shifts are uncorrelated with industry Xs, controlling for period FE
- Uncorrelated with regional Xs, controlling for period FExlagged manuf. share

| Balance variable | Coef. | SE |
|---|---|---|
| Panel A: Industry-level balance | | |
| Production workers' share of employment, 1991 | −0.011 | (0.012) |
| Ratio of capital to value-added, 1991 | −0.007 | (0.019) |
| Log real wage (2007 USD), 1991 | −0.005 | (0.022) |
| Computer investment as share of total, 1990 | 0.750 | (0.465) |
| High-tech equipment as share of total investment, 1990 | 0.532 | (0.296) |
| No. of industry-periods | 794 | |
| Panel B: Regional balance | | |
| Start-of-period % of college-educated population | 0.915 | (1.196) |
| Start-of-period % of foreign-born population | 2.920 | (0.952) |
| Start-of-period % of employment among women | −0.159 | (0.521) |
| Start-of-period % of employment in routine occupations | −0.302 | (0.272) |
| Start-of-period average offshorability index of occupations | 0.087 | (0.075) |
| Manufacturing employment growth, 1970s | 0.543 | (0.227) |
| Manufacturing employment growth, 1980s | 0.055 | (0.187) |
| No. of region-periods | 1,444 | |

# The China Shock: Estimates/SE from Industry-year Reg.

- China shock is larger in the 2000s (post WTO entry)
- This is when overall manufactaring decline is stronger for other reasons

| | (1) | (2) | (3) | (4) | (5) | (6) | (7) |
|---|---|---|---|---|---|---|---|
| Coefficient | −0.596 | −0.489 | −0.267 | −0.314 | −0.310 | −0.290 | −0.432 |
| | (0.114) | (0.100) | (0.099) | (0.107) | (0.134) | (0.129) | (0.205) |
| Regional controls | | | | | | | |
| Autor *et al.* (2013) controls | ✓ | ✓ | ✓ | | ✓ | ✓ | ✓ |
| Start-of-period mfg. share | ✓ | | | | | | |
| Lagged mfg. share | | ✓ | ✓ | ✓ | ✓ | ✓ | ✓ |
| Period-specific lagged mfg. share | | | ✓ | ✓ | ✓ | ✓ | ✓ |
| Lagged 10-sector shares | | | | | ✓ | | ✓ |
| Local Acemoglu *et al.* (2016) controls | | | | | | ✓ | |
| Lagged industry shares | | | | | | | ✓ |
| SSIV first stage $F$-stat. | 185.6 | 166.7 | 123.6 | 272.4 | 64.6 | 63.3 | 27.6 |
| No. of region-periods | 1,444 | 1,444 | 1,444 | 1,444 | 1,444 | 1,444 | 1,444 |
| No. of industry-periods | 796 | 794 | 794 | 794 | 794 | 794 | 794 |

# SSIV as combining diff-in-diffs

# Special Case 1: One Time Period, Two Industries

- $S_{i2} = 1 - S_{i1}$
- $Z_i = S_{i1}g_1 + S_{i2}g_2 = S_{i1}g_1 + (1 - S_{i1})g_2 = g_2 + (g_1 - g_2)S_{i1}$
- First Stage

$$D_i = \gamma_0 + \gamma Z_i + \eta_i$$
$$D_i = \underbrace{\gamma_0 + \gamma g_2}_{\text{constant}} + \underbrace{\gamma(g_1 - g_2)}_{\text{coefficient}} S_{i1} + \eta_i$$

- The instrument is $S_{i1}$, while $g_k$ affects relevance

# Special Case 2: T Periods, Two Industries

- $Z_{it} = S_{i10} g_{1t} + S_{i20} g_{2t} = g_{2t} + (\Delta_{gt}) S_{i10}$
- First Stage

$$D_i = \tau_i + \tau_t + \gamma Z_{it} + \eta_{it}$$
$$D_i = \tau_i + \underbrace{\tau_t + \gamma g_{2t}}_{\tilde{\tau}_t} + \underbrace{\gamma \Delta_{gt}}_{\tilde{\gamma}_t} S_{i10} + \eta_{it}$$

- Industry shares $\times$ time period is the instrument: $Z_{it} = \underbrace{\Delta_{gt}}_{\text{size}} \underbrace{S_{i10}}_{\text{exposure}}$
- Analogy to continuous diff-in-diff: use "pre-period" to test // pre-trends

# General Case; Pooling Diff-in-Diffs

- Assume exogenous shares: Cov $[\epsilon_i, S_{ik}] = 0$ for every $k$
- With $Y_i$ measured in differences, this is $K$ parallel trend assumptions
- Shares are measured in the pre-period
- Rules out any unobserved shocks $\nu_k$ that affect $Y_i$ via $S_{ik}$ (or similar shares)
  - $\Rightarrow$ Even if shares are drawn at random, these shocks would violate // trends
- Then we have $K$ valid IVs: $S_{i1}, ..., S_{iK}$
  - $\Rightarrow$ SSIV with $Z_i = \sum_k S_{ik} g_k$ is one reasonable way to combine them
  - $\Rightarrow$ 2SLS (for small $K$, or even one $k$) is another alternative

# Plausible Vs. Implausible Exogenous Shares

- Construct the instrument with shares that are tailored to the treatment
- Mediating only the shocks to $D_{it}$ and not other shocks that might affect $Y_{it}$
- E.g. Immigration shares: local migration from various origins in the past
- E.g. industry employment shares are generic: reflect exposure to other shocks

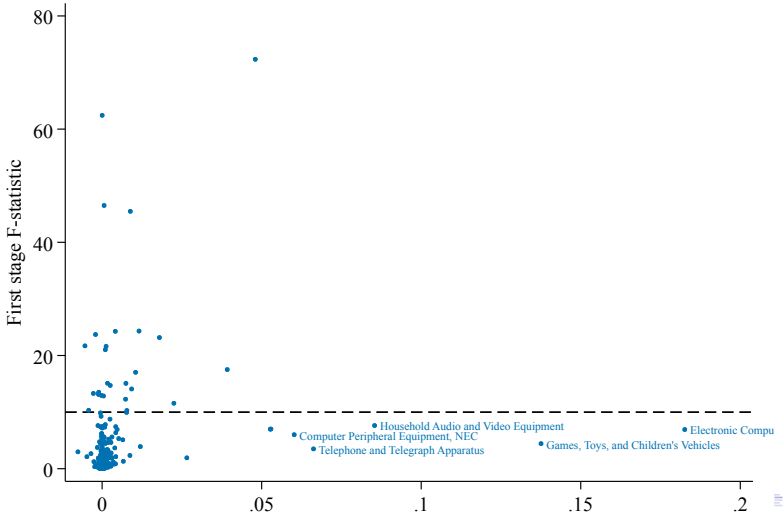
$\Rightarrow$ Same instrument for different treatments $\rightarrow$ problematic

# Rotemberg Weights

- $\hat{\tau} = \sum_k \hat{\alpha}_k \hat{\tau}_k$ for $\hat{\tau}_k$ that uses $S_{ik}$ as IV one at a time
- $\hat{\alpha}_k$ are higher for $k$ with more extreme shifts and larger first stages
- $\hat{\alpha}_k$ add up to one but can be negative (potential issue w/heterog. effects)
- Scrutinize validity of the share IVs with highest (Rotemberg) weights

$\Rightarrow$ Sensitivity of the SSIV estimate to violations of exogeneity by each share IV

# China Shock Example (cont'ed): Rotemberg Weights



# Summary of Main Practical Takeaways

| | Approach | |
|---|---|---|
| | **Many exogenous shifts** (1) | **Exogenous shares** (2) |
| **Identification argument** | Shifts are as-good-as-randomly assigned and only affect the outcome through the treatment | Each share satisfies parallel trends: the outcomes of units with high vs. low shares would have trended the same if not for the treatment |
| **Estimation** | Control for the sum of shares (if not one) and shift-share aggregates of any shift-level controls | Check robustness to using share instruments directly: e.g., one share at a time or pooled via two-stage least squares or limited information maximum likelihood |
| **Statistical inference** | Get exposure-robust standard errors from the equivalent shift-level instrumental variable regression | Use conventional heteroskedasticity- or cluster-robust standard errors |
| **Balance tests** | For both the shift-share instrument and the shifts | For both the shift-share instrument and the shares with high Rotemberg weights |
| **Do not use when...** | You would not want to use the shifts directly as an instrument in a shift-level regression, e.g. because they are too few or endogenous | You would not want to use shares directly as instruments, e.g. because they are "generic" (capturing the unit's exposure to many types of shocks) |

# Rural migration and Urban Informality: Evidence from Brazil

# Imbert and Ulyssea (2025)

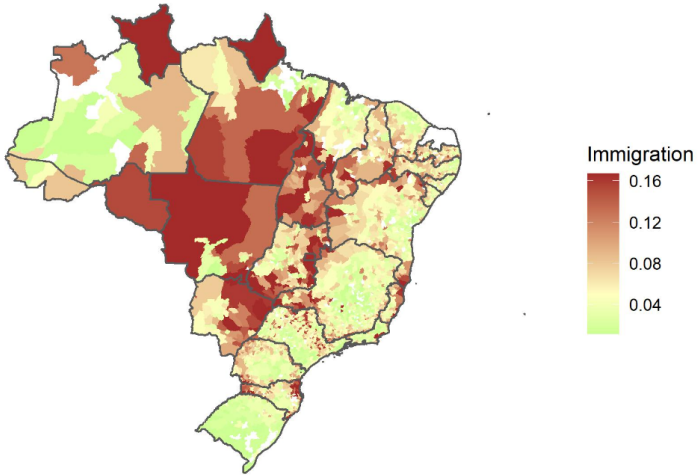
- Study the labor market effects of rural-urban migration on Brazilian cities
- SSIV to identify the causal effects of immigration at destination
  - $\Rightarrow$ $\uparrow$ formal employees, $\downarrow$ informal employees and $\downarrow$ wages
  - $\Rightarrow$ $\uparrow$ formal firms, $\uparrow$ jobs, $\uparrow$ entry, and $\uparrow$ exit
- Model of firms dynamics with inf. and heterog. growth profiles across firms

$\Rightarrow$ Eq. effects of migration with more/less enforc., with/without wage rigidity

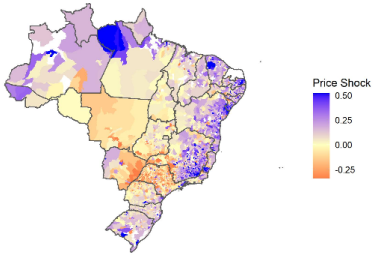
# Data

- Migration and Labor Market Outcomes: Population Census, 1991-2010
  - $\Rightarrow$ Migrants = in their current location $\leq$ 10 years
  - $\Rightarrow$ Focus migration to urban areas (88%), and cross-state borders
- Firms' data
  - $\Rightarrow$ Matched employer-employee, universe of formal firms and workers
  - $\Rightarrow$ Survey data on small (up to 5 employees) formal and informal firms
- Push shocks
  - $\Rightarrow$ Int. agri. commodity price shocks  crop shares at the muni. level (in 1980)
  - $\Rightarrow$ Drought index  growing season by crop  crop shares

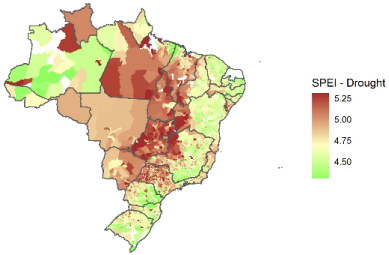
# Immigration, 2000-2010



# Migration Push Shocks




(a) Price Shocks




(b) Drought Shocks

# Shift-Share Design

$$\Delta y_d = \beta_0 + \beta_1 Mig_d + \beta_2' X_d + u_d$$

- $\Delta y_d = y_{d,2010} - y_{d,2000}$ (Census or RAIS)
- $Mig_d = \sum_o \sum_{t=2001}^{2010} \frac{Mig_{o,d,t}}{Pop_{d,2000}}$
- $X_d$: share of male, young, and high skill (HS completed) in 2000

$\Rightarrow$ Instrument: $Z_d = \sum_o \underbrace{\lambda_{o,d}}_{\text{shares}} \underbrace{s_o^{\text{drought}}}_{\text{shift}}$

# Shift-Share Design (cont'd)

- Baseline migration shares may be endogenous to future outcomes

$\Rightarrow$ Push-shocks (shifts) are as good as randomly assigned

$\Rightarrow$ Instrument only affects the dependent variables via the immigration shares

- A few specification/robustness checks
  - Pre-trends: dynamic effects + lagged changes in outcomes as a control
  - Persistence of migration (the shares): control for lagged migration rates
  - Capital Channel: Control for exposure through bank network
  - Demand Channel: Control for local price shocks and shocks to neighboring regions weighted by distance
  - Second push instrument coming from variations in international crop prices

# Effects of Immigration on Workers

| | Wage employment | | | Log monthly wage | | |
|---|---|---|---|---|---|---|
| | Overall (1) | Formal (2) | Informal (3) | Overall (4) | Formal (5) | Informal (6) |
| Panel A: OLS | | | | | | |
| Immigration | 0.037* | 0.105*** | −0.068*** | 0.062 | 0.032 | 0.035 |
| | (0.019) | (0.023) | (0.014) | (0.076) | (0.068) | (0.092) |
| Panel B: IV-Drought | | | | | | |
| Immigration | −0.014 | 0.271*** | −0.284*** | −0.126 | −0.671** | −0.200 |
| | (0.060) | (0.089) | (0.072) | (0.284) | (0.336) | (0.352) |
| F Statistic (IV) | 18.11 | 18.11 | 18.11 | 18.11 | 18.11 | 18.11 |
| Baseline Mean | 0.332 | 0.229 | 0.103 | - | - | - |
| Observations | 3,545 | 3,545 | 3,545 | 3,545 | 3,545 | 3,545 |

Notes: Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality's population in 2000.

# Effects of Immigration on Firms

| | Nb firms (1) | Entry (2) | Exit (3) | Nb jobs (4) | Firm wage (5) |
|---|---|---|---|---|---|
| Panel A: OLS | | | | | |
| Immigration | 1.342*** | 1.167*** | 1.588*** | 1.071*** | 0.372*** |
| | (0.108) | (0.203) | (0.274) | (0.270) | (0.101) |
| Panel B: IV-Drought | | | | | |
| Immigration | 1.625*** | 2.555*** | 2.758** | 2.031*** | −0.747 |
| | (0.306) | (0.877) | (1.123) | (0.624) | (0.554) |
| F Statistic (IV) | 18.11 | 18.11 | 18.11 | 18.11 | 18.11 |
| Observations | 3,545 | 3,545 | 3,545 | 3,545 | 3,545 |

Notes: Robust standard errors in parenthesis. All regressions control for the share of women, youth (less than 18 years old), and high skill individuals (at least completed high-school) measured in 2000. All regressions are weighted by municipality's population in 2000. Columns 1-3 refer to the log of total number of firms, entrants and exiting firms, respectively.

# Shift-Share Design: Summary of Results

- 1p.p. $\uparrow$ in $Mig_d(14.5\%SD) \rightarrow \uparrow$ 0.4 p.p. in share of formal $\approx$ 2.1% increase

$\Rightarrow$ Formalization effect driven by workers moving from informal to formal jobs

- 1p.p. $\uparrow$ in $Mig_d \rightarrow \downarrow$ formal wages by 0.7%, no effect on informal wages
- 1p.p. $\uparrow$ in $Mig_d \rightarrow \uparrow$ 2.4% in the number of firms, 2.2% in nb of formal jobs

$\Rightarrow$ There is greater churn, effect on entry is slightly higher

# Model: Overview

- Continuum of firms indexed by their individual productivity $\theta$
- Same technology, labor only input: $f(\ell) = \theta q(\ell), q' > 0, q'' < 0$
- Tradeoffs are driven by regulations and enforcement
  - $\Rightarrow$ Informal firms: lower entry costs, and no regulation costs (e.g. taxes), but cost of operation is increasing in firm size
  - $\Rightarrow$ Formal firms: face all regulatory costs, but constant marginal costs; can evade labor regulations by hiring informal workers
- Endogenous exit + exogenous death shock (different across sectors)
- No aggregate shocks, homogenous labor, and labor supply is fixed

# Firm Profits

- Informal firms avoid taxes but pay an informality cost

$$\Pi_i(\theta, w) = \max_{\ell}\{\theta q(\ell) - \tau_i(\ell)w\}, \quad \tau_i' > 0, \tau_i'' > 0$$

- Formal firms pay revenue taxes but can evade payroll tax

$$\Pi_f(\theta, w) = \max_{\ell}\{(1-\tau_y)\theta q(l) - C(\ell)\}$$

$$C(l) = \begin{cases} \tau_f(\ell)w & \text{for } \ell \leq \tilde{\ell} \\ \tau_f(\tilde{\ell})w + (1+\tau_w)(\ell - \tilde{\ell}) & \text{for } \ell > \tilde{\ell}, \end{cases}$$

- Where $\tau_f' > 0, \tau_f'' > 0$ and $\tilde{\ell}$ such that $\tau_f'(\tilde{\ell}) = 1 + \tau_w$

# Firm Dynamics

- Dynamics is driven by the evolution of firms' idiosyncratic productivity $\theta$
- Productivity process is given by

$$\begin{aligned}\ln \theta_{j,1} &= \ln \nu_j + \ln \epsilon_{j,1} \\ \ln \theta_{j,t} &= \rho_s \ln \theta_{j,t-1} + (1-\rho_s) \ln \nu_j + \ln \epsilon_{j,t}, \quad t \geq 2\end{aligned}$$

- Where $j$ indexes firms, $s \in \{i, f\}$ denotes sector, and $\ln \epsilon \sim \ln N(0, \sigma_s^2)$
- Formal firms cannot become informal
- Informal firms can pay $\tilde{c}^e = c_f^e - c_i^e$ and formalize

# Firm Dynamics (cont'd)

- The value functions of formal and informal incumbents

$$\begin{aligned} V_f(\theta, w) =& \tau_f(\theta, w) + (1-\delta_f)\beta \max\{0, E_\nu[V_f(\theta', w) \mid \theta]\} \\ V_i(\theta, w) =& \tau_i(\theta, w) \\ &+\beta \max\{0, (1-\delta_i)E_\nu[V_i(\theta', w) \mid \theta], (i-\delta_f)E_\nu[V_f(\theta', w) \mid \theta] - \tilde{c}^e\} \end{aligned}$$

- where $\beta$ is the discount factor, $\delta_s$ the exogenous exit

- Exit decisions and informal-to-formal transitions follow cutoff rules

$$\begin{aligned} E_\nu[V_s(\theta', w) \mid \underline{\theta}_s] &= 0, \quad s \in \{i, f\} \\ E_\nu[V_f(\theta', w) - V_i(\theta', w) \mid \overline{\theta}_i] &= \tilde{c}^e \end{aligned}$$

# Firm Entry

- Entrants in both sectors pay a fixed cost of entry, denoted by $c_s^e, s \in \{i, f\}$
- These parameters will be estimated, but theory suggests that $c_f^e > c_i^e$
- Expected value of entry for firm with long-run productivity $\nu$: $E[V_s(\theta, w) \mid \nu]$
- Entry is also characterized by cutoff rules

$$E_\nu[V_i(\theta, w) \mid \underline{\nu}_i] = c_i^e$$
$$E_\nu[V_f(\theta, w) - V_i(\theta, w) \mid \underline{\nu}_f] = c_f^e - c_i^e$$

- Where $\underline{\nu}_s$ characterizes the last firm to enter sector $s = i, f$

# Empirical Predictions

- Immigration shock: ↑ both labor supply and demand for the final good
- Lower real wages and higher labor demand for incumbent firms
- Incumbents are limited by scale, so ↑ in labor supply → greater firm entry
- In eq. not only effect on entry rates but also exit rates and firm size distrib.
- Net effect on informality, or output per worker, is unclear
  - ⇒ If migrants are largely absorbed by informal firms and/or low-productivity formal firms: ↑ informality among firms and workers
  - ⇒ if lower wages induce more entry then more firm growth in the formal sector and lower informality

# Model Parametrization

- Production function: $y(\theta, \ell) = \theta \ell^{\alpha}, \quad 0 < \alpha < 1$
- Cost functions: $\tau_s(\ell) = \left(1 + \frac{\ell}{\varphi_s}\right) \ell, \quad \varphi_s > 0$

$\Rightarrow$ The larger $\varphi_i$ the more informal firms can grow, the larger $\varphi_f$ the easier for formal firms to hire informal workers

- Long-run productivity $\nu \sim H$, observed before entry occurs

$$H(\nu \geq x) = \begin{cases} \left(\frac{\nu_0}{x}\right)^{\xi} & \text{for } x \geq \nu_0 \\ 1 & \text{for } x < \nu_0 \end{cases}$$

$\Rightarrow$ First productivity draw $\theta_1 = \nu \epsilon_1$ has a Pareto-Lognormal distribution

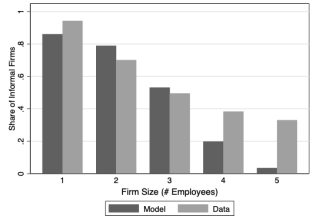
# Estimation

- Two step SMM procedure (set $\tau_w, \tau_y$ to statutory values)
  - $\Rightarrow$ First step: system GMM and panel data on productivity process $\rho_f$
  - $\Rightarrow$ Second step: 12 remaining parameter estimated using SMM
- Moments
  - $\Rightarrow$ Overall share of informal firms and by size brackets
  - $\Rightarrow$ Average share of informal workers within formal firms
  - $\Rightarrow$ Formal and informal firm growth at ages 5 and 10 relative to age 1
  - $\Rightarrow$ Formal/informal firms size distribution by size brackets

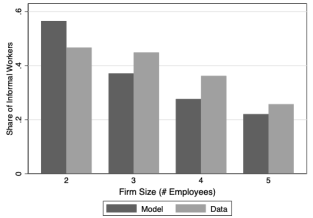
# Model's parameters

| Parameter | Description | Source | Value |
|---|---|---|---|
| *First Step* | | | |
| $\tau_w$ | Payroll Tax | Statutory values | 0.375 |
| $\tau_y$ | Revenue Tax | Statutory values | 0.293 |
| $\rho$ | Productivity Process: Persistence Parameter | GMM Estimation | 0.92 |
| $\nu_0$ | Pareto's Location Parameter | Calibrated | 7.3 |
| $\gamma_f$ | Per-period fixed cost (Formal) | Calibrated | 0.7 |
| *Second Step: Min. Dist. Calibration* | | | |
| $\varphi_f$ | Intensive margin: $\tau_f = \left(1 + \frac{\ell}{\varphi_f}\right) \ell$ | – | 5.830 |
| $\varphi_i$ | Extensive margin: $\tau_i = \left(1 + \frac{\ell}{\varphi_i}\right) \ell$ | – | 5.427 |
| $\delta_i$ | Informal death shock | – | 0.148 |
| $\delta_f$ | Formal death shock | – | 0.066 |
| $\gamma_i$ | Per-period fixed cost (Informal) | – | 0.340 |
| $\xi$ | Pareto shape parameter | – | 3.801 |
| $c_f^{e\,\dagger}$ | Formal sector's entry cost | – | 6,205 |
| $c_i^{e\,\dagger}$ | Informal sector's entry cost | – | 2,800 |
| $\alpha$ | Span-of-control | – | 0.643 |
| $\sigma_i$ | Informal productivity process: SD | – | 0.144 |
| $\sigma_f$ | Formal productivity process: SD | – | 0.145 |
| $\rho_i$ | Informal productivity process: persistence | – | 0.935 |

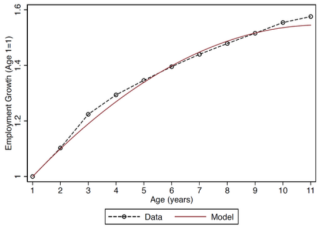
# Model Fit: Non-targeted Moments



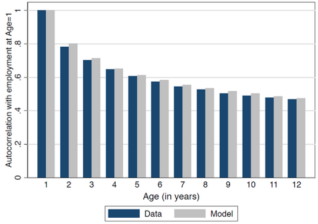

(a) Extensive Mg.




(b) Intensive Mg.




(c) Formal Firms' growth



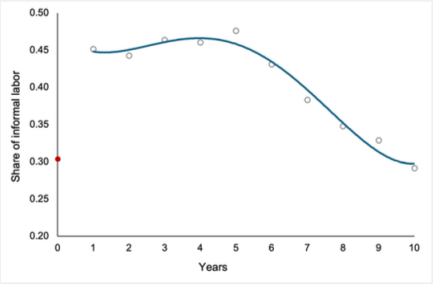

(d) Log-Employment Autocorrelations

# Counterfactuals

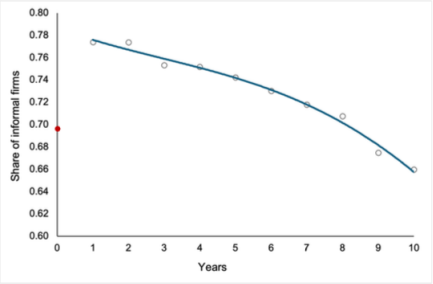
| | IV Estimation | Labor Supply Shock | LS Shock + Enforcement |
|---|---|---|---|
| Share Informal Workers (Δ% ) | -7.3 | -4.3 | -38.2 |
| Wages (Δ% ) | -5.3 | -3.4 | -2.1 |
| Number Formal Firms (Δ% ) | 20.7 | 16.3 | 128 |
| Newly created firms | – | 9.9 | – |
| Previously informal firms | – | 6.4 | – |
| Share Informal Firms (Δ% ) | – | -5.3 | -68.2 |
| Average Firm Productivity (Δ% ) | – | -1.4 | 2.5 |
| Output (Δ% ) | – | 7.1 | 8.3 |
| Taxes (Δ% ) | – | 8.7 | 30.9 |

Notes: IV estimation results from a regression contrasting municipalities below and above the median immigration rate (see text). Labor supply shock corresponds to simulating a permanent increase of 10% in population, and effects are measured as percentage change relative to baseline values.

# From Short- to Long-Run: Wage Rigidity in the Formal Sector




(a) Labor Informality




(b) Firm Informality

Notes: Red dot indicates the initial steady state of the baseline economy.

# Main Takeways from the Estimated Model

- Model revisits empirical evidence on the LMK effect of (internal) immigration
- $\downarrow$ wages, $\uparrow$ entry of formal firms, nb of formal firms, jobs and formal. share

$\Rightarrow$ Gains do not accrue to the most productive firms $\rightarrow$ output per worker falls

$\Rightarrow$ Despite its role as a "stepping-stone" in the short run, the informal sector reduces the aggregate benefits of immigration

$\Rightarrow$ The typical Harris-Todaro-Fields result arises with downward wage rigidity in the formal sector: higher informality + lower output gains

# SSIV $\Leftrightarrow$ Economic Model

- Model attempts to rationalize surprising results from SSIV
- The model quantitatively replicates the long-run IV results
- Model sheds light on mechanism for $\uparrow$ in formal firms (40% comes from the formalization of informal firms, and 60% comes from the creation of new formal firms)

$\Rightarrow$ Firm dynamics and the linkages between the informal and formal sectors